

APPOINTMENT

University of Connecticut

Assistant Professor, Department of Electrical and Computer Engineering

Storrs, CT, USA

2025–Present

Columbia University in the City of New York

Postdoctoral Research Scientist, Columbia Nano Initiative

New York, NY, USA

2021–2025

EDUCATION

University of California, Santa Barbara

Ph.D. in Electrical and Computer Engineering

Santa Barbara, CA, USA

2018–2021

University of California, Santa Barbara

M.S. in Electrical and Computer Engineering

Santa Barbara, CA, USA

2015–2018

Tsinghua University

B.Eng. in Electronic Engineering

Beijing, China

2011–2015

PUBLICATION

Book Chapters

- B1 Y. Wang, X. Meng, M. Lipson, A. L. Gaeta, and K. Bergman, “Co-packaged optics (CPO) for communications,” in *Optical Fiber Telecommunications VIII*, A. E. Willner, Ed., chapter manuscript with the volume editor; book publication scheduled for March 1, 2027. Elsevier, 2027

Journal Articles

- J1 Z. Wu, Y. Wang, and K. Bergman, “SiPAM: Silicon Photonic Accelerated Memory Pooling Architecture,” *IEEE Micro*, pp. 1–10, 2026. doi 10.1109/MM.2026.3670009
- J2 R. Parsons, K. Jang, Y. Wang, A. Novick, A. M. Smith, C. C. Tison, Y. Gebregiorgis, V. Deenadayalan, M. van Niekerk, L. Carpenter, T. Ngai, G. Leake, D. Coleman, X. Meng, S. Preble, M. L. Fanto, K. Bergman, and A. Rizzo, “Highly uniform thermally undercut silicon photonic devices in a 300 mm CMOS foundry process,” *Scientific Reports*, vol. 15, no. 1, p. 29 906, Aug. 2025. doi 10.1038/s41598-025-14480-4
- J3 R. Parsons, A. Oh, J. Robinson, S. Wang, M. Cullen, K. Jang, A. James, Y. Wang, and K. Bergman, “Foundry-enabled wafer-scale characterization and modeling of silicon photonic DWDM links,” *Nanophotonics*, vol. 14, no. 27, pp. 5363–5374, 2025. doi 10.1515/nanoph-2025-0439
- J4 J. Robinson, R. Parsons, Y. Wang, K. Jang, X. Meng, and K. Bergman, “Silicon photonic DWDM micro-resonator link initialization under fabrication variation,” *Optics Express*, vol. 33, no. 26, pp. 53 700–53 713, Dec. 2025. doi 10.1364/OE.577064
- J5 S. Wang, Y. Wang, S. Sanyal, R. Parsons, X. Ji, Y. Okawachi, X. Meng, M. Lipson, A. Gaeta, and K. Bergman, “Automated tuning of a ring-assisted MZI-based interleaver for Kerr frequency combs,” *Optics Letters*, vol. 50, no. 2, p. 698, Jan. 2025. doi 10.1364/OL.546772
- J6 Y. Wang, S. Wang, R. Parsons, S. Sanyal, V. Gopal, A. Novick, A. Rizzo, M. Lipson, A. L. Gaeta, and K. Bergman, “Co-Designed Silicon Photonics Chip I/O for Energy-Efficient Petascale Connectivity,” *IEEE Transactions on Components, Packaging and Manufacturing Technology*, vol. 15, no. 8, pp. 1581–1591, Aug. 2025. doi 10.1109/TCPMT.2024.3492189 2025 IEEE Transactions on Components, Packaging and Manufacturing Technology Best Paper Award recipient
- J7 Z. Wu, L. Yuan Dai, Y. Wang, S. Wang, and K. Bergman, “Flexible silicon photonic architecture for accelerating distributed deep learning,” *Journal of Optical Communications and Networking*, vol. 16, no. 2, A157, Feb. 2024. doi 10.1364/JOCN.497372
- J8 A. James, A. Rizzo, Y. Wang, A. Novick, S. Wang, R. Parsons, K. Jang, M. Hattink, and K. Bergman, “Process Variation-Aware Compact Model of Strip Waveguides for Photonic Circuit Simulation,” *Journal of Lightwave Technology*, pp. 1–14, 2023. doi 10.1109/JLT.2023.3238847
- J9 A. Novick, A. James, L. Y. Dai, Z. Wu, A. Rizzo, S. Wang, Y. Wang, M. Hattink, V. Gopal, K. Jang, R. Parsons, and K. Bergman, “High-bandwidth density silicon photonic resonators for energy-efficient optical interconnects,” *Applied Physics Reviews*, vol. 10, no. 4, p. 041 306, Dec. 2023. doi 10.1063/5.0160441
- J10 Y. Wang, P. Sun, J. Hulme, M. A. Seyed, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, “Energy Efficiency and Yield Optimization for Optical Interconnects via Transceiver Grouping,” *Journal of Lightwave Technology*, vol. 39, no. 6, pp. 1567–1578, Mar. 2021. doi 10.1109/JLT.2020.3039489

- J11 Z. Zhang, R. Wu, **Y. Wang**, C. Zhang, E. J. Stanton, C. L. Schow, K.-T. Cheng, and J. E. Bowers, "Compact Modeling for Silicon Photonic Heterogeneously Integrated Circuits," *Journal of Lightwave Technology*, vol. 35, no. 14, pp. 2973–2980, Jul. 2017. doi [10.1109/JLT.2017.2706721](https://doi.org/10.1109/JLT.2017.2706721)

Conference Proceedings

- C1 M. Cullen, S. Sanyal, S. Wang, K. J. McNulty, **Y. Wang**, M. Lipson, A. L. Gaeta, and K. Bergman, "Reconfigurable Kerr-comb for enhanced link margins in >1 Tb/s direct-drive DWDM links," in *Conference on Lasers and Electro-Optics (CLEO)*, Charlotte, NC: Optica Publishing Group, May 2026, STh4A.1. doi [10.1364/CLEO_SI.2026.STh4A.1](https://doi.org/10.1364/CLEO_SI.2026.STh4A.1) Highlighted
- C2 B. C. George, W. Wang, Z. Wu, **Y. Wang**, X. Meng, M. Ghobadi, and K. Bergman, "Reconfiguration-aware direct-connect AI cluster using spatial-and-wavelength selective switching," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2026, M3F.4. doi [10.1364/OFC.2026.M3F.4](https://doi.org/10.1364/OFC.2026.M3F.4) Top-Scored
- C3 S. Sanyal, M. Cullen, S. Wang, **Y. Wang**, Y. Okawachi, K. J. McNulty, M. Lipson, K. Bergman, and A. L. Gaeta, "High-power Kerr comb source for data communications," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2026, Th1F.5. doi [10.1364/OFC.2026.Th1F.5](https://doi.org/10.1364/OFC.2026.Th1F.5)
- C4 S. Wang, M. Cullen, T. Zypman, S. Sanyal, K. J. McNulty, **Y. Wang**, X. Meng, M. Lipson, A. L. Gaeta, and K. Bergman, "On-chip programmable MZI-based Fourier synthesizer for ultra-broadband Kerr-comb equalization," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2026, M1E.4. doi [10.1364/OFC.2026.M1E.4](https://doi.org/10.1364/OFC.2026.M1E.4)
- C5 N. Nauman, J. Robinson, **Y. Wang**, K. Jang, X. Meng, and K. Bergman, "Photonic Analog-to-Digital Architecture for Accelerating Multiply-Accumulate Operations," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2025, W3D.5. doi [10.1364/OFC.2025.W3D.5](https://doi.org/10.1364/OFC.2025.W3D.5)
- C6 R. Parsons, S. Sanyal, M. Cullen, **Y. Wang**, A. Novick, X. Ji, Y. Okawachi, X. Meng, M. Lipson, A. Gaeta, and K. Bergman, "Dispersion-Engineered Resonator-Based Interleaver Co-Designed with Kerr Comb Source," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2025, Tu2G.6. doi [10.1364/OFC.2025.Tu2G.6](https://doi.org/10.1364/OFC.2025.Tu2G.6)
- C7 A. Rovinski, Y. Ou, C. Ou, D. Khilwani, **Y. Wang**, S. Wang, S. Lee, K. Bergman, A. Molnar, and C. Batten, "Scaling Co-Packaged Optical Interconnects Using Hybrid 2.5D/3D Integration," in *IEEE International Symposium on Circuits and Systems (ISCAS)*, May 2025, pp. 1–5. doi [10.1109/ISCAS56072.2025.11043946](https://doi.org/10.1109/ISCAS56072.2025.11043946)
- C8 **Y. Wang**, S. Wang, S. Sanyal, N. Nauman, R. Parsons, J. Robinson, M. Hattink, K. Jang, A. Novick, K. J. McNulty, X. Meng, M. Lipson, A. L. Gaeta, and K. Bergman, "Order-Preserving Channel Calibration of Kerr Comb-Driven Microresonator-Based DWDM Link," in *Optical Fiber Communication Conference (OFC)*, Optica Publishing Group, Mar. 2025, W1E.1. doi [10.1364/OFC.2025.W1E.1](https://doi.org/10.1364/OFC.2025.W1E.1)
- C9 A. Novick, M. Hattink, A. Rizzo, **Y. Wang**, V. Gopal, S. Wang, R. Parsons, and K. Bergman, "Integrated Photonic Resonant Modulator-Based Equalization and Optimization for DWDM," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2024, W1K.5. doi [10.1364/OFC.2024.W1K.5](https://doi.org/10.1364/OFC.2024.W1K.5)
- C10 S. Wang, **Y. Wang**, X. Meng, K. Hosseini, T. T. Hoang, and K. Bergman, "Automated Tuning of Ring-Assisted MZI-Based Interleaver for DWDM Systems," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2024, Th1A.3. doi [10.1364/OFC.2024.Th1A.3](https://doi.org/10.1364/OFC.2024.Th1A.3)
- C11 **Y. Wang**, S. Wang, R. Parsons, A. Novick, V. Gopal, K. Jang, A. Rizzo, C.-P. Chiu, K. Hosseini, T. T. Hoang, S. Shumarayev, and K. Bergman, "Silicon Photonics Chip I/O for Ultra High-Bandwidth and Energy-Efficient Die-to-Die Connectivity," in *IEEE Custom Integrated Circuits Conference (CICC)*, Denver, CO, USA: IEEE, Apr. 2024, pp. 1–8. doi [10.1109/CICC60959.2024.10529018](https://doi.org/10.1109/CICC60959.2024.10529018)
- C12 Z. Wu, R. Parsons, S. Wang, **Y. Wang**, and K. Bergman, "Wavelength Reconfigurable Transceiver for Multi-Interface Compute Accelerator Networks," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2024, W4F.2. doi [10.1364/OFC.2024.W4F.2](https://doi.org/10.1364/OFC.2024.W4F.2)
- C13 G. Michelogiannakis, Y. Arafa, B. Cook, L. Y. Dai, A.-H. Hameed Badawy, M. Glick, **Y. Wang**, K. Bergman, and J. Shalf, "Efficient Intra-Rack Resource Disaggregation for HPC Using Co-Packaged DWDM Photonics," in *IEEE International Conference on Cluster Computing (CLUSTER)*, Santa Fe, NM, USA: IEEE, Oct. 2023, pp. 158–172. doi [10.1109/CLUSTER52292.2023.00021](https://doi.org/10.1109/CLUSTER52292.2023.00021)
- C14 S. Wang, A. Novick, A. Rizzo, R. Parsons, S. Sanyal, K. J. McNulty, B. Y. Kim, Y. Okawachi, **Y. Wang**, A. Gaeta, M. Lipson, and K. Bergman, "Integrated, Compact, and Tunable Band-Interleaving of a Kerr Comb Source," in *Conference on Lasers and Electro-Optics (CLEO)*, San Jose, CA: Optica Publishing Group, 2023, STh3J.6. doi [10.1364/CLEO_SI.2023.STh3J.6](https://doi.org/10.1364/CLEO_SI.2023.STh3J.6)
- C15 **Y. Wang**, A. Novick, R. Parsons, S. Wang, K. Jang, A. James, M. Hattink, V. Gopal, A. Rizzo, C.-P. Chiu, K. Hosseini, T. T. Hoang, and K. Bergman, "Scalable architecture for sub-pJ/b multi-Tbps comb-driven DWDM silicon photonic transceiver," in *Next-Generation Optical Communication: Components, Sub-Systems, and Systems XII*, G. Li, K. Nakajima, and A. K. Srivastava, Eds., San Francisco, United States: SPIE, Mar. 2023, p. 55. doi [10.1117/12.2649506](https://doi.org/10.1117/12.2649506)
- C16 **Y. Wang**, S. Wang, A. Novick, A. James, R. Parsons, A. Rizzo, and K. Bergman, "Dispersion-Engineered and Fabrication-Robust SOI Waveguides for Ultra-Broadband DWDM," in *Optical Fiber Communication Conference (OFC)*, San Diego, CA, USA: Optica Publishing Group, 2023, Th3A.4. doi [10.1364/OFC.2023.Th3A.4](https://doi.org/10.1364/OFC.2023.Th3A.4)
- C17 A. James, **Y. Wang**, A. Rizzo, and K. Bergman, "Flexible, Process-Aware Compact Model of Effective Index in Silicon Waveguides for Commercial Foundries," in *International Conference on Numerical Simulation of Optoelectronic Devices (NUSOD)*, Turin, Italy: IEEE, Sep. 2022, pp. 173–174. doi [10.1109/NUSOD54938.2022.9894784](https://doi.org/10.1109/NUSOD54938.2022.9894784)

- C18 **Y. Wang** and K.-T. Cheng, "Traffic-Adaptive Power Reconfiguration for Energy-Efficient and Energy-Proportional Optical Interconnects," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, Munich, Germany: IEEE, Nov. 2021, pp. 1–9. doi: [10.1109/ICCAD51958.2021.9643475](https://doi.org/10.1109/ICCAD51958.2021.9643475)
- C19 **Y. Wang**, J. Hulme, P. Sun, M. Jain, M. A. Seyed, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Characterization and Applications of Spatial Variation Models for Silicon Microring-Based Optical Transceivers," in *ACM/IEEE Design Automation Conference (DAC)*, San Francisco, CA, USA: IEEE, Jul. 2020, pp. 1–6. doi: [10.1109/DAC18072.2020.9218608](https://doi.org/10.1109/DAC18072.2020.9218608)
- C20 **Y. Wang** and K.-T. Cheng, "Task Mapping-Assisted Laser Power Scaling for Optical Network-on-Chips," in *IEEE/ACM International Conference on Computer-Aided Design (ICCAD)*, Westminster, CO, USA: IEEE, Nov. 2019, pp. 1–6. doi: [10.1109/ICCAD45719.2019.8942146](https://doi.org/10.1109/ICCAD45719.2019.8942146)
- C21 **Y. Wang**, M. A. Seyed, J. Hulme, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Bidirectional tuning of microring-based silicon photonic transceivers for optimal energy efficiency," in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Tokyo, Japan: ACM, Jan. 2019, pp. 370–375. doi: [10.1145/3287624.3287649](https://doi.org/10.1145/3287624.3287649)
- C22 **Y. Wang**, L. Shao, M. A. Lastras-Montano, and K.-T. Cheng, "Taming Emerging Devices' Variation and Reliability Challenges with Architectural and System Solutions [Invited]," in *IEEE International Conference on Microelectronic Test Structures (ICMTS)*, Kita-Kyushu City, Fukuoka, Japan: IEEE, Mar. 2019, pp. 90–95. doi: [10.1109/ICMTS.2019.8730924](https://doi.org/10.1109/ICMTS.2019.8730924)
- C23 **Y. Wang**, M. A. Seyed, R. Wu, J. Hulme, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Energy-efficient channel alignment of DWDM silicon photonic transceivers," in *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Dresden, Germany: IEEE, Mar. 2018, pp. 601–604. doi: [10.23919/DATE.2018.8342079](https://doi.org/10.23919/DATE.2018.8342079)
- C24 R. Wu, M. A. Seyed, **Y. Wang**, J. Hulme, M. Fiorentino, R. G. Beausoleil, and K.-T. Cheng, "Pairing of microring-based silicon photonic transceivers for tuning power optimization," in *Asia and South Pacific Design Automation Conference (ASP-DAC)*, Jeju, South Korea: IEEE, Jan. 2018, pp. 135–140. doi: [10.1109/ASPDAC.2018.8297295](https://doi.org/10.1109/ASPDAC.2018.8297295)
- C25 R. Wu, **Y. Wang**, Z. Zhang, C. Zhang, C. L. Schow, J. E. Bowers, and K.-T. Cheng, "Compact modeling and circuit-level simulation of silicon nanophotonic interconnects," in *Design, Automation & Test in Europe Conference & Exhibition (DATE)*, Lausanne, Switzerland: IEEE, Mar. 2017, pp. 602–605. doi: [10.23919/DATE.2017.7927057](https://doi.org/10.23919/DATE.2017.7927057)
- C26 A. Ghofrani, M. A. Lastras-Montano, **Y. Wang**, and K.-T. Cheng, "In-place Repair for Resistive Memories Utilizing Complementary Resistive Switches," in *International Symposium on Low Power Electronics and Design (ISLPED)*, San Francisco Airport, CA, USA: ACM, Aug. 2016, pp. 350–355. doi: [10.1145/2934583.2934590](https://doi.org/10.1145/2934583.2934590)
- C27 C. Xu, F. X. Lin, **Y. Wang**, and L. Zhong, "Automated OS-level Device Runtime Power Management," in *International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, Istanbul, Turkey: ACM, Mar. 2015, pp. 239–252. doi: [10.1145/2694344.2694360](https://doi.org/10.1145/2694344.2694360)

Under Review

- U1 B. George, **Y. Wang**, W. Wang, Z. Wu, X. Meng, M. Ghobadi, and K. Bergman, "RADICAL: Reconfiguration-Aware Direct-Connect AI Cluster using Spatial-and-Wavelength Selective Switching," *IEEE/Optica Journal of Optical Communications and Networking*, 2026

MENTORING

U : undergraduate; **M** : master's; **D** : doctoral

Waseem Faheem D Ph.D. Student, University of Connecticut	Spring 2026 – Present
Zahin Zeal D Ph.D. Student, University of Connecticut	Fall 2026 – Present
Harvansh Pelia U Undergraduate Student, University of Connecticut <i>Honors Thesis Supervision</i>	Spring 2026 – Present
Willow Pichardo U Undergraduate Student, University of Connecticut	Spring 2026 – Present
Uyen Bao Pham U Undergraduate Student, University of Connecticut	Spring 2026 – Present
Khang "James" Le U Undergraduate Student, University of Connecticut	Spring 2026 – Present
Abidur Rahman U Undergraduate Student, Wayne State University <i>Summer Undergraduate Research Experience (SURE) Program of Columbia Engineering</i>	Summer 2024

TEACHING

U : undergraduate; G : graduate

University of Connecticut

Storrs, CT, USA

- Instructor ECE 2001: Electrical Circuits U Spring 2026
- Project Advisor ECE 4901: Senior Design U Spring 2026

Columbia University in the City of New York

New York, NY, USA

- Guest Lecturer ELEN 9404: Seminar in Lightwave Communications G Spring 2024
- Guest Lecturer ELEN 9403: Seminar in Photonics G Spring 2023

University of California, Santa Barbara

Santa Barbara, CA, USA

- Teaching Assistant ECE 153B: Sensor & Peripheral Interface Design U Winter 2019

SERVICE

Guest Editor IEEE Solid-State Circuits Magazine 2027
Spring 2027 Issue: Silicon Photonics

Technical Program Committee Optical Fiber Communication Conference (OFC) 2027
N2: Optics and Photonics for Data Center and Computing Applications

Topical Chair IEEE Photonics Society Summer Topicals Meeting Series (SUM) 2026
Electronic-Photonic Systems at Scale — Architecture, Methodology, and Going 3D (EP3D)

Technical Program Committee IEEE Hot Interconnects Symposium (HotI) 2026

Technical Program Committee IEEE Chiplet Workshop: Standards, Circuits and Systems 2025
Chiplet Interconnects and Packaging

Ad-hoc Reviewer

- *Nature Photonics*, Springer Nature
- *Nature Nanotechnology*, Springer Nature
- *Nature Communications*, Springer Nature
- *npj Nanophotonics*, Springer Nature
- *Communications Engineering*, Springer Nature
- *APL Photonics*, American Institute of Physics
- *Photonics*, MDPI
- *ACM Transactions on Design Automation of Electronic Systems*, ACM
- *Journal of Lightwave Technology*, IEEE/Optica
- *Journal of Selected Areas in Communications*, IEEE
- *Journal on Emerging and Selected Topics in Circuits and Systems*, IEEE
- *IEEE Transactions on Computers*, IEEE
- *IEEE Transactions on Computer-Aided Design of Integrated Circuits and Systems*, IEEE
- *IEEE Transactions on Very Large Scale Integration (VLSI) Systems*, IEEE
- *IEEE Access*, IEEE

AWARD AND HONOR

PRESENTATION

- Seminar** Department of Physics, University of Connecticut Storrs, CT, USA
Integrated Photonics Systems in the Package: Beyond Massive Parallel Data Movement Mar. 2026
- Seminar** Department of ECE, University of Connecticut Storrs, CT, USA
Photonics-Empowered Computing: From Efficient to Functional Data Movement Feb. 2025
- Seminar** School of ECEE, Arizona State University Tempe, AZ, USA
Photonics-Empowered Computing: From Efficient to Functional Data Movement Feb. 2025
- Seminar** Department of ECE, University of Minnesota Minneapolis, MN, USA
Photonics-Empowered Computing: From Efficient to Functional Data Movement Feb. 2025
- Presentation** SRC TECHCON Austin, TX, USA
Scalable Kerr-Comb Driven DWDM Silicon Photonics Chip I/O Sep. 2024
- Seminar** School of EECS, Pennsylvania State University University Park, PA, USA
Co-Designing Photonics for Heterogeneously Integrated Systems Apr. 2024
- Invited Talk** SPIE Photonics West San Francisco, CA, USA
Scalable Architecture for Sub-pJ/b Multi-Tbps Comb-Driven DWDM Silicon Photonic Transceiver Jan. 2023
- Poster** Ph.D. Forum, ACM/IEEE Design Automation Conference (DAC) Online Virtual Event
Design and Optimization of Variation-Aware Runtime-Reconfigurable Optical Interconnects Jun. 2020
- Invited Talk** Optical/Photonic Interconnects for Computing Systems (OPTICS) Workshop Dresden, Germany
Optimal Pairing and Non-Uniform Channel Alignment of Microring-Based Transceivers for Comb Laser-Driven DWDM Silicon Photonics Mar. 2018
- Seminar** Department of ECE, Hong Kong University of Science and Technology Hong Kong SAR, China
Variation-Aware Modeling and Design of Silicon Photonic Systems Jan. 2018
- Poster** Optical/Photonic Interconnects for Computing Systems (OPTICS) Workshop Lausanne, Switzerland
Variation-Aware Modeling and Design of Nanophotonic Interconnects Mar. 2017